

[Your University Name]

Department of Electrical Engineering / Electronics &
Telecommunications

Applications of Electronic Circuits in Communication Engineering

Project Title:

Beamforming Optimization for IRS-Aided Wireless Systems
Using Particle Swarm Optimization

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Abstract

This report presents the study and implementation of beamforming optimization for an Intelligent Reflecting Surface (IRS) aided wireless communication system. We first review the practical phase shift model proposed by Abeywickrama *et al.* [1], which captures the amplitude-phase coupling in IRS reflecting elements – a critical hardware imperfection ignored by most prior work. We then apply **Particle Swarm Optimization (PSO)** to jointly optimize the transmit beamforming at the access point and the IRS phase shifts, with the goal of maximizing the achievable spectral efficiency. Simulation results are compared against the Alternating Optimization (AO) baseline reported in the original paper. Our results show that **[TODO: summarize your key finding here after running simulations]**.

Contents

1	Introduction	3
1.1	Motivation	3
1.2	Problem Statement	3
1.3	Chosen Algorithm	3
1.4	Report Organization	3
2	System Model	4
2.1	Network Topology	4
2.2	Channel Model	4
2.3	Signal Model	4
2.4	Achievable Rate	5
2.5	Practical Phase Shift Model	5
3	Optimization Algorithm: Particle Swarm Optimization	5
3.1	Basic Description	5
3.2	Update Rules	6
3.3	Pseudocode	6
3.4	Computational Complexity	7
3.5	Convergence Behavior	7
3.6	Variants and Parameter Settings	7
4	Problem Formulation and Algorithm Application	7
4.1	Joint Beamforming Optimization Problem	7
4.2	Decoupling via MRT Beamforming	8
4.3	PSO Applied to IRS Phase Shift Optimization	8

5	Simulation Results	8
5.1	Simulation Setup	8
5.2	Achievable Rate vs. AP-User Distance	9
5.3	Achievable Rate vs. Number of IRS Elements	9
5.4	Convergence of PSO	9
5.5	Discussion	9
6	Conclusion	10

1 Introduction

1.1 Motivation

Wireless communication systems constantly demand higher data rates, better coverage, and improved energy efficiency. Traditional solutions – such as deploying more base stations or using relay nodes – are costly and power-hungry. *Intelligent Reflecting Surfaces* (IRS) have recently emerged as a low-cost, passive alternative that can reshape the wireless propagation environment without requiring active radio-frequency components.

An IRS consists of a large number of passive reflecting elements, each capable of independently adjusting the phase (and to some extent the amplitude) of the reflected signal. By coordinating these elements, an IRS can steer the reflected signals toward a desired receiver, effectively creating a constructive interference channel even in heavily obstructed environments.

1.2 Problem Statement

Most existing IRS research assumes an *ideal phase shift model*, where each element reflects with unity amplitude regardless of its phase shift. However, as demonstrated in [1], practical IRS elements exhibit a strong coupling between amplitude and phase shift due to inherent hardware losses. Designing beamforming strategies under this ideal assumption leads to measurable performance degradation in real systems.

This project addresses the joint optimization of:

1. **Transmit beamforming** at the access point (AP), and
2. **IRS phase shifts**, under the practical phase shift model,

to maximize the achievable rate at a single-antenna user.

1.3 Chosen Algorithm

To solve this non-convex optimization problem, we employ **Particle Swarm Optimization (PSO)**, a population-based metaheuristic algorithm inspired by the collective behavior of bird flocks.

1.4 Report Organization

The remainder of this report is organized as follows. Section 2 describes the system model and the practical phase shift model. Section 3 presents the chosen optimization algorithm in detail. Section 4 formulates the optimization problem and explains how the algorithm

is applied. Section 5 presents and discusses simulation results. Section 6 concludes the report.

2 System Model

2.1 Network Topology

We consider a Multiple-Input Single-Output (MISO) downlink system consisting of:

- An **Access Point (AP)** equipped with M antennas,
- An **IRS** composed of N passive reflecting elements,
- A **single-antenna user**.

The system geometry is illustrated in Figure 1. The IRS is deployed between the AP and user to assist communication, particularly when the direct AP-user link is weak due to obstacles.

[TODO: Insert system diagram here – e.g., exported from Python/draw.io]

Figure 1: IRS-aided MISO wireless communication system.

2.2 Channel Model

Three channels are defined:

- $\mathbf{h}_d \in \mathbb{C}^{M \times 1}$: the direct channel from AP to user,
- $\mathbf{h}_r \in \mathbb{C}^{N \times 1}$: the channel from IRS to user,
- $\mathbf{G} \in \mathbb{C}^{N \times M}$: the channel from AP to IRS.

All channels follow a quasi-static flat-fading model and are assumed known at the AP. Path loss is modeled with a reference attenuation of 40 dB at 1 m, and path loss exponents of 2.2 (AP-IRS), 2.8 (IRS-user), and 3.8 (AP-user) [1].

2.3 Signal Model

The AP transmits signal $x = \mathbf{w}s$, where $\mathbf{w} \in \mathbb{C}^{M \times 1}$ is the beamforming vector and s is the data symbol with $\mathbb{E}[|s|^2] = 1$. The received signal at the user is:

$$y = (\mathbf{v}^H \mathbf{\Phi} + \mathbf{h}_d^H) \mathbf{w}s + z, \quad (1)$$

where $\mathbf{\Phi} = \text{diag}(\mathbf{h}_r^H) \mathbf{G}$ combines the AP-to-IRS and IRS-to-user channels, $\mathbf{v} \in \mathbb{C}^{N \times 1}$ is the IRS reflection coefficient vector, and $z \sim \mathcal{CN}(0, \sigma^2)$ is additive white Gaussian noise.

2.4 Achievable Rate

The achievable spectral efficiency (bits/s/Hz) is:

$$R = \log_2 \left(1 + \frac{|(\mathbf{v}^H \mathbf{\Phi} + \mathbf{h}_d^H) \mathbf{w}|^2}{\sigma^2} \right). \quad (2)$$

Maximizing R is equivalent to maximizing the received SNR.

2.5 Practical Phase Shift Model

Unlike the ideal model which assumes $|v_n| = 1$ for all phase shifts, the practical model [1] captures the hardware reality that amplitude and phase are coupled. The n -th element's reflection coefficient is:

$$v_n = \beta_n(\theta_n) e^{j\theta_n}, \quad \theta_n \in [-\pi, \pi), \quad (3)$$

where the amplitude $\beta_n(\theta_n)$ is given by:

$$\beta_n(\theta_n) = (1 - \beta_{\min}) \left(\frac{\sin(\theta_n - \phi) + 1}{2} \right)^k + \beta_{\min}. \quad (4)$$

The parameters $\beta_{\min}$, ϕ , and k characterize the hardware:

- $\beta_{\min} \geq 0$: the minimum reflection amplitude (floor),
- $\phi \geq 0$: horizontal offset of the amplitude minimum,
- $k \geq 0$: controls the sharpness of the amplitude dip. When $k = 0$, (4) reduces to the ideal model ($\beta_n = 1$).

In our simulations, we use $\beta_{\min} = 0.2$, $k = 1.6$, and $\phi = 0.43\pi$, as reported in [1].

Remark 1. *The minimum amplitude occurs near $\theta_n = 0$ and the maximum near $\theta_n = \pm\pi$. This means simply setting $\theta_n = \angle\varphi_n$ (as in the ideal design) is suboptimal when $\angle\varphi_n \approx 0$.*

3 Optimization Algorithm: Particle Swarm Optimization

3.1 Basic Description

Particle Swarm Optimization (PSO) [3] is a population-based metaheuristic that simulates the social behavior of a flock of birds searching for food. Each candidate solution is represented as a *particle* moving through the N -dimensional search space of IRS phase shifts.

Each particle i maintains:

- $\mathbf{x}_i$: its current position (a candidate phase shift vector),
- $\mathbf{p}_i$: its personal best position found so far,
- $\mathbf{g}$: the global best position found by any particle,
- $\mathbf{v}_i^{\text{vel}}$: its velocity (direction of movement).

3.2 Update Rules

At each iteration t , velocities and positions are updated as:

$$\mathbf{v}_i^{(t+1)} = w \mathbf{v}_i^{(t)} + c_1 r_1 (\mathbf{p}_i - \mathbf{x}_i^{(t)}) + c_2 r_2 (\mathbf{g} - \mathbf{x}_i^{(t)}), \quad (5)$$

$$\mathbf{x}_i^{(t+1)} = \mathbf{x}_i^{(t)} + \mathbf{v}_i^{(t+1)}, \quad (6)$$

where w is the inertia weight, c_1 and c_2 are cognitive and social coefficients, and $r_1, r_2 \sim \mathcal{U}(0, 1)$ are random scalars.

3.3 Pseudocode

Algorithm 1 PSO for IRS Phase Shift Optimization

Require: N (IRS elements), S (swarm size), T (max iterations), channels $\mathbf{h}_d, \mathbf{h}_r, \mathbf{G}$, model params $\beta_{\min}, \phi, k$

Ensure: Optimal phase shifts θ^* , maximum rate R^*

```

1: Initialize: for each particle  $i = 1, \dots, S$ :  $\mathbf{x}_i \leftarrow \text{Uniform}(-\pi, \pi)^N$ ,  $\mathbf{v}_i^{\text{vel}} \leftarrow \mathbf{0}$ ,  $\mathbf{p}_i \leftarrow \mathbf{x}_i$ 
2:  $\mathbf{g} \leftarrow \arg \max_i f(\mathbf{x}_i)$ 
3: for  $t = 1$  to  $T$  do
4:   for  $i = 1$  to  $S$  do
5:     Compute  $\beta_n$  from (4) for each element  $n$ 
6:     Build reflection vector:  $v_n = \beta_n e^{jx_{i,n}}$ 
7:     Compute combined channel:  $\mathbf{c} = \mathbf{v}^H \Phi + \mathbf{h}_d^H$ 
8:     Compute MRT beamforming:  $\mathbf{w}^* = \sqrt{P_T} \mathbf{c}^H / \|\mathbf{c}\|$ 
9:     Evaluate fitness:  $f_i = \log_2(1 + |\mathbf{c} \mathbf{w}^*|^2 / \sigma^2)$ 
10:    if  $f_i > f(\mathbf{p}_i)$  then  $\mathbf{p}_i \leftarrow \mathbf{x}_i$ 
11:    end if
12:    if  $f_i > f(\mathbf{g})$  then  $\mathbf{g} \leftarrow \mathbf{x}_i$ 
13:    end if
14:    Update velocity via (5)
15:    Update position via (6)
16:    Clamp:  $\mathbf{x}_i \leftarrow \text{clip}(\mathbf{x}_i, -\pi, \pi)$ 
17:  end for
18: end for
19: return  $\mathbf{g}, f(\mathbf{g})$ 

```

3.4 Computational Complexity

The per-iteration cost is $\mathcal{O}(S \cdot N \cdot M)$, where S is the swarm size, N the number of IRS elements, and M the number of AP antennas. Total complexity over T iterations is $\mathcal{O}(T \cdot S \cdot N \cdot M)$. For typical values ($T = 200$, $S = 30$, $N = 40$, $M = 2$), this is computationally tractable on a standard laptop.

3.5 Convergence Behavior

PSO is not guaranteed to find the global optimum due to the non-convex nature of the problem. However, empirically it converges within **[TODO: X]** iterations for this problem size (see Figure 4). The inertia weight w is linearly decayed from 0.9 to 0.4 over iterations to balance exploration (early) and exploitation (late).

3.6 Variants and Parameter Settings

[TODO: Briefly mention 1-2 PSO variants, e.g., LPSO (local topology), APSO (adaptive PSO), or PSO with constriction factor. Then state the exact hyperparameters you used.]

The hyperparameters used in this work are summarized in Table 1.

Table 1: PSO hyperparameters used in simulation.

Parameter	Symbol	Value
Swarm size	S	30
Max iterations	T	200
Inertia weight	w	$0.9 \rightarrow 0.4$ (linear decay)
Cognitive coeff.	c_1	2.0
Social coeff.	c_2	2.0

4 Problem Formulation and Algorithm Application

4.1 Joint Beamforming Optimization Problem

The joint optimization of AP beamforming $\mathbf{w}$ and IRS phase shifts $\{\theta_n\}$ is formulated as:

$$\max_{\mathbf{w}, \mathbf{v}, \{\theta_n\}} \left| \left(\mathbf{v}^H \mathbf{\Phi} + \mathbf{h}_d^H \right) \mathbf{w} \right|^2 \quad \text{s.t.} \quad \|\mathbf{w}\|^2 \leq P_T, \quad v_n = \beta_n(\theta_n) e^{j\theta_n}, \quad \theta_n \in [-\pi, \pi). \quad (7)$$

4.2 Decoupling via MRT Beamforming

For any fixed IRS phase shift vector $\mathbf{v}$, the optimal transmit beamforming is Maximum Ratio Transmission (MRT):

$$\mathbf{w}^* = \sqrt{P_T} \frac{(\mathbf{v}^H \Phi + \mathbf{h}_d^H)^H}{\|\mathbf{v}^H \Phi + \mathbf{h}_d^H\|}. \quad (8)$$

Substituting (8) into (7) reduces the problem to optimizing only the IRS phase shifts $\{\theta_n\}$, which PSO handles directly.

4.3 PSO Applied to IRS Phase Shift Optimization

Each PSO particle encodes a candidate phase shift vector $\mathbf{x}_i = [\theta_1, \dots, \theta_N] \in [-\pi, \pi)^N$. The fitness function evaluated for each particle is the achievable rate R in (2), computed using the practical model (4) and the MRT beamforming (8).

PSO searches the N -dimensional continuous space for the phase configuration that maximizes R , without requiring gradient information or convexity assumptions.

5 Simulation Results

5.1 Simulation Setup

The simulation parameters follow those of [1] exactly, to allow direct comparison:

Table 2: Simulation parameters.

Parameter	Symbol	Value
AP antennas	M	2
IRS elements	N	40 (Fig. 5), varied (Fig. 6)
AP-IRS distance		500 m
Reference path loss		40 dB at 1 m
Path loss exp. (AP-IRS)	α_1	2.2
Path loss exp. (IRS-User)	α_2	2.8
Path loss exp. (AP-User)	α_3	3.8
Transmit power	P_T	36 dBm
Noise variance	σ^2	−94 dBm
Phase shift model params	$\beta_{\min}, k, \phi$	0.2, 1.6, 0.43π
Channel realizations		1000

5.2 Achievable Rate vs. AP-User Distance

[TODO: Insert your reproduced Figure 5 here]

Figure 2: Achievable rate vs. AP-user horizontal distance ($N = 40$). Comparison of: (1) Ideal IRS upper bound, (2) PSO with practical model, (3) AO with practical model (paper baseline), (4) Ideal model assumption on practical hardware, (5) No IRS lower bound.

[TODO: Discuss your Figure 2 here. Compare your PSO curve against the AO curves from the paper. Comment on: (a) where PSO matches AO, (b) where it diverges and why, (c) the performance gap between practical and ideal model.]

5.3 Achievable Rate vs. Number of IRS Elements

[TODO: Insert your reproduced Figure 6 here]

Figure 3: Achievable rate vs. number of reflecting elements ($d = 498$ m).

[TODO: Discuss Figure 3: how does performance scale with N ? Does the practical model gap grow with N ? Why?]

5.4 Convergence of PSO

[TODO: Insert convergence plot here: best fitness vs. iteration number]

Figure 4: PSO convergence behavior for a single channel realization ($N = 40$, $d = 498$ m).

[TODO: Comment on the number of iterations needed for convergence and the effect of swarm size on convergence speed.]

5.5 Discussion

[TODO: Write 1-2 paragraphs summarizing:

- How well does PSO reproduce the paper's results?
- What are PSO's advantages over AO for this problem (e.g., no closed-form derivation needed, handles discrete phases)?
- What are its disadvantages (e.g., slower convergence, hyperparameter sensitivity)?

]

6 Conclusion

This report studied the beamforming optimization problem for an IRS-aided MISO wireless communication system under the practical phase shift model proposed in [1]. The key contributions of this work are:

1. A summary of the IRS system model and the practical phase shift model, highlighting the amplitude-phase coupling that invalidates the ideal assumption.
2. A detailed description and implementation of Particle Swarm Optimization for solving the joint transmit and reflect beamforming problem.
3. Simulation results that **[TODO: describe your finding – e.g., “confirm that PSO achieves within X% of the AO baseline while requiring no gradient computation”]**.

Future work could explore PSO variants with adaptive inertia, hybrid PSO-AO approaches, or extension to multi-user scenarios.

References

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- [4] C. Huang, A. Zappone, G. C. Alexandropoulos, M. Debbah, and C. Yuen, “Reconfigurable intelligent surfaces for energy efficiency in wireless communication,” *IEEE Trans. Wireless Commun.*, vol. 18, no. 8, pp. 4157–4170, Aug. 2019.
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